BYTEBITE: SMART HEALTHY MEAL PLANNER & BUDGET LOCATOR APP

SDG Innovation Challenge: From Problem to Sustainable Solution (Lean Canvas Final Output)

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Market Size: TAM 600,000 | SAM 25,000 | SOM 1,000 Subscribers | Service Area: Cubao, Quezon City

SDGs Addressed: SDG 3 (Good Health & Well-Being), SDG 2 (Zero Hunger), SDG 1 (No Poverty), SDG 12

STEP 1: IDENTIFY AN SDG PROBLEM & PROBLEM STATEMENT

Primary SDGs Addressed: SDG 3 (Good Health & Well-Being - Preventing CKD/Dialysis, Diabetes & Hypertension), SDG 2 (Zero Hunger), SDG 1 (No Poverty), and SDG 12 (Responsible Consumption).

3-5 Sentence Problem Statement (Answering All 5 Required Questions):

Across urban university corridors in the Philippines such as Cubao, Quezon City, there is a high and rising prevalence of preventable lifestyle diseases-specifically Chronic Kidney Disease (CKD) requiring dialysis, hypertension, and diabetes-among young adults. University students and entry-level workers on tight budgets (PHP 45 to PHP 150 per meal) and busy schedules rely heavily on cheap, high-sodium fast food, instant noodles, and canned goods packing 1,500mg-1,900mg of sodium per plate because they lack a tool to plan affordable healthy meals or locate healthy local karinderyas. According to the National Kidney and Transplant Institute (NKTI), over 1 in 10 Filipinos suffer from kidney disease, making diet-induced renal failure both a life-threatening crisis for young people and a catastrophic financial burden for families. Addressing this urgent problem requires a pure digital smart meal planner and budget karinderya locator aligned with UN SDG 3 (Good Health and Well-Being), SDG 2 (Zero Hunger), and SDG 1 (No Poverty).

- 1. What is the problem? High prevalence of lifestyle diseases (Chronic Kidney Disease/Dialysis, Diabetes, Hypertension) caused by reliance on cheap, high-sodium fast food and instant meals.
- 2. Who is affected? College students (e.g., TIP Quezon City) and young urban adults managing busy schedules and tight PHP 45-PHP 235 meal budgets.
- 3. Where does the problem occur? Cubao, Quezon City (SAM: 25,000 students/young adults across TIP QC, Araneta City, and Aurora Blvd; TAM: 600,000 across Metro Manila).
- 4. Why does it need to be solved? To prevent irreversible kidney failure (dialysis), hypertension, and diabetes while helping students eat healthy within their daily allowance.
- 5. What SDG does it support? SDG 3 (Good Health & Well-Being), SDG 2 (Zero Hunger), SDG 1 (No Poverty), and SDG 12 (Responsible Consumption).

STEP 2: DEVELOP YOUR SOLUTION (PRODUCT/SERVICE CONCEPT)

Product/Service Name: ByteBite: Smart Healthy Meal Planner & Budget Locator App (Pure Digital App for PC & Mobile)

SDG Addressed:SDG 3 (Good Health & Well-Being), SDG 2 (Zero Hunger), SDG 1 (No Poverty), SDG 12 (Responsible Consumption)

Target User: University students and young adults in Cubao, Quezon City (SAM: 25,000; Target SOM: 1,000 active subscribers).

One-Sentence Description: ByteBite is a pure digital app that generates customized,

kidney-safe healthy meal plans based on student budgets (Low PHP 45-75, Medium PHP 80-135, High PHP 140-235), scans meals in real time via AI camera, and instantly locates affordable healthy vendors and karinderyas in Cubao, QC.

Proposed Solution Explanation (6 Sentences):

ByteBite is a pure digital web and mobile application that empowers busy students and young adults to prevent lifestyle diseases (Chronic Kidney Disease/dialysis, diabetes, and hypertension) without exceeding their daily food allowance. First, the app generates customized, healthy Filipino meal plans categorized into real Philippine Peso tiers: Low Tier (PHP 45-75 carinderia/dorm savers), Medium Tier (PHP 80-135 campus/office plates), and High Tier (PHP 140-235 high-protein bowls), complete with itemized palengke ingredient costs and 12 audio-narrated Cooking Reels. Second, ByteBite features an interactive Live GPS Map (Leaflet + OpenStreetMap) that instantly locates 6 verified, affordable healthy vendors and karinderyas in Cubao, Quezon City with real-time Haversine distance routing. Third, its Real-Time AI Camera & Barcode Scanner (powered by TensorFlow.js MobileNet) lets users scan what they are eating on the spot, immediately flagging high-sodium dialysis-risk foods (like instant pancit canton or fast-food fried chicken) and suggesting a PHP 48 Ginisang Monggo healthy swap. Fourth, the dedicated Kidney & Sodium Shield tracks daily sodium against the 1,500mg-2,000mg DOH/WHO limit alongside a daily water-flush hydration calculator. Engineered with an adaptive PC Holder and Mobile Phone Holder interface and priced at PHP 99.00/month for ByteBite Pro (with a PHP 80.00 contribution margin per user), ByteBite reaches break-even at just 313 subscribers and generates PHP 55,000.00 in monthly net profit at our 1,000-user SOM target.

STEP 3: COMPLETE THE LEAN CANVAS (ALL 9 BLOCKS)

1. Problem (Top 3 Problems)

- 1. High Prevalence of Lifestyle Diseases: Rising rates of Chronic Kidney Disease (CKD/Dialysis), Diabetes, and Hypertension among young adults due to excessive sodium (3,500mg-5,000mg/day) and processed diets.
- 2. Reliance on Cheap High-Sodium Fast Food: Students rely on instant noodles, canned meats, and fast food due to busy class schedules and tight daily budgets (PHP 45-150).
- 3. Lack of Local Healthy Karinderya Visibility: Students struggle to find nearby Cubao vendors/karinderyas serving freshly cooked, low-sodium vegetable and fish meals at student-friendly prices.

2. Customer Segments & Market Sizing (TAM, SAM, SOM)

- Primary Customers: University students (TIP Quezon City, Cubao student belt) and young urban adults managing tight food budgets and busy schedules.
- End Users: Daily student meal planners, dormitory cooks, and health-conscious mobile/PC app users.
- Beneficiaries: Local Cubao healthy karinderyas/vendors gaining digital foot traffic and Filipino families spared from costly dialysis treatments.
- Market Sizing: TAM (Total Addressable Market) = 600,000 Metro Manila students/young adults | SAM (Serviceable Available Market) = 25,000 in Cubao, QC | SOM (Serviceable Obtainable Market) = 1,000 active paid subscribers/month.

3. Unique Value Proposition (UVP)

Completed Statement: "Our product/service helps university students and young adults in Cubao, QC achieve customized, kidney-safe healthy nutrition within their real PHP 45 to PHP 235 budget by providing a pure digital smart meal planner, real-time AI food scanner, and instant GPS locator for affordable healthy karinderyas."

Differentiation: Unlike generic Western calorie trackers or expensive food delivery apps that add 30% markups, ByteBite is a pure digital app hyper-localized to Cubao karinderyas,

real Philippine palengke prices (Low/Medium/High tiers), and DOH kidney-safe sodium thresholds.

4. Solution (Main Features & Functions)

- Customized Budget Meal Planner (1,009+ Meals): Generates healthy Filipino meal plans across Low (PHP 45-75), Medium (PHP 80-135), and High (PHP 140-235) tiers + 7-Day Pro Schedule.
- Instant Cubao Healthy Karinderya GPS Locator: Interactive Leaflet/OpenStreetMap map pinning 6 verified low-sodium vendors in Cubao, QC with real GPS distance and directions.
- Real-Time AI Camera & Barcode Scanner: Scans meals live via webcam/phone camera to display calories, protein, PHP cost, and Kidney Dialysis Risk alerts.
- Kidney & Sodium Shield + 12 Audio Cooking Reels: Monitors daily sodium (<1,500mg-2,000mg), hydration flush, and step-by-step palengke recipe videos.

5. Channels

- Pure Digital Cloud Deployment (Vercel / GitHub / Public HTTPS Link & QR Code): Instant access on any PC Holder or Mobile Phone Holder browser.
- Campus & Student Organization Partnerships: University clinics, student councils, and GEE/PE wellness campaigns at TIP Quezon City.
- Partner Karinderya Digital Referrals: In-store QR codes at partner eateries around Cubao, Aurora Blvd, and Araneta City.
- Social Media Short-Form Video: Budget healthy cooking reels on TikTok, Facebook Reels, and campus groups.

6. Revenue Streams

- Monthly Subscription Fee (ByteBite Pro): PHP 99.00 / user per month.
- Target Monthly Volume (SOM): 1,000 active paid subscribers generating PHP 99,000.00 in Monthly Revenue.
- Contribution Margin: PHP 80.00 / user (80.8% contribution margin after PHP 19.00 variable cost per user).

7. Cost Structure & Unit Economics

- Variable Cost per Subscription: PHP 19.00 / user (Cloud API queries, messaging support, payment gateway fees) -> Total Variable Cost at 1,000 users = PHP 19,000.00 / month.
- Monthly Fixed Costs: PHP 25,000.00 / month (Cloud hosting, developer retainer, campus marketing, admin fees).
- Break-Even Volume (BEP): 313 paid subscribers/month (PHP 25,000 Fixed Costs / PHP 80 Contribution Margin = 312.5 -> 313 subscribers).

8. Key Metrics (5 Measurable Indicators)

- 1. Active Paid Subscribers (SOM Progress): Reaching 313 subscribers for Break-Even and 1,000 active subscribers for PHP 99,000/mo revenue.
- 2. Daily Sodium & Lifestyle Risk Reduction: Keeping user daily sodium intake below the 1,500mg-2,000mg DOH/WHO kidney-safe limit.
- 3. Student Food Budget Savings: Saving users PHP 60-PHP 110/day by locating PHP 45-PHP 75 healthy Cubao karinderya meals.
- 4. Karinderya Locator Directions & Redemptions: Monthly GPS map clicks and partner vendor visits in Cubao.
- 5. Net Monthly Profit & Contribution Margin: Maintaining PHP 80.00/user contribution margin and PHP 55,000.00 monthly net profit at 1,000 users.

9. Unfair Advantage

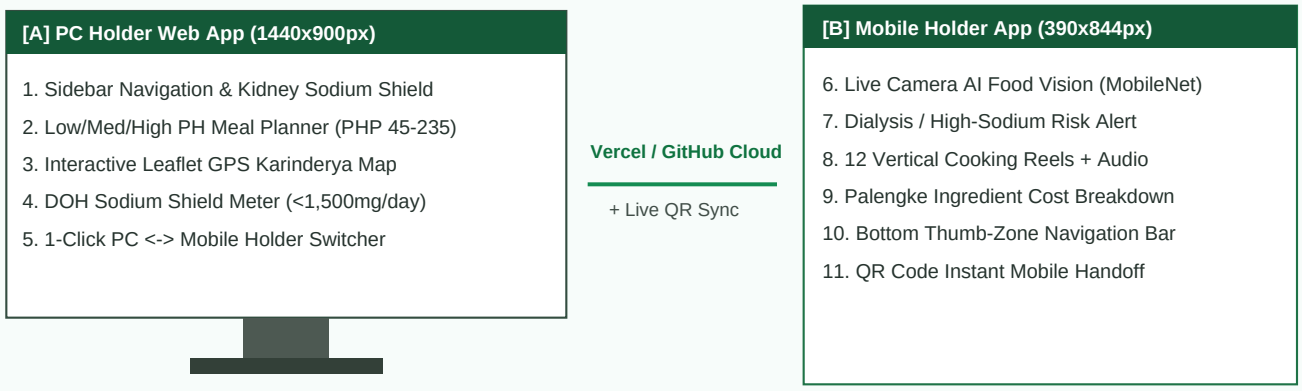
Hyper-localized Cubao Karinderya & Palengke Nutrition Database (1,009+ Filipino meals mapped to Low/Medium/High PHP budgets and DOH kidney sodium thresholds) combined with a pure-digital, ultra-low variable cost architecture (PHP 19/user variable cost via

client-side TensorFlow.js Edge AI and OpenStreetMap).

STEP 4: PRODUCT DESIGN, LABELED SKETCH & TECHNICAL SPECIFICATIONS

A. Product Name: ByteBite: Smart Healthy Meal Planner & Budget Locator App (Pure Digital Dual-Holder Web & Mobile App)

FIG 1. LABELED PURE-DIGITAL APP SKETCH (PC HOLDER WORKSTATION + MOBILE PHONE HOLDER)



Pure Digital App Specifications: Mobile Holder 390x844px | PC Holder 1440x900px | Capacity: 1,009+ Meals, 6 Cubao Karinderyas

C. Basic Dimensions, Capacity & Weight (Digital UI & Cloud Specs)

- Mobile Holder Viewport Dimensions: Width 390 px (fits 71.5 mm x 146.7 mm smartphones) x Height 844 px with 6-icon thumb navigation bar.
- PC Holder Viewport Dimensions: Width 1440 px x Height 900 px widescreen desktop layout with 248 px command sidebar.
- System Capacity & Footprint: 1,009+ Filipino meals, 52 real dish photos, 12 WebM cooking videos, 6 GPS Cubao karinderyas; Pure Digital App (0 g physical weight; <35 MB cloud bundle).

D. Materials & Technology Stack

- Frontend & Mobile Holder UI: HTML5, CSS3 Responsive Dual-Holder Design System, Vanilla ES2022 JavaScript.
- AI & Geospatial Mapping: TensorFlow.js 4.17 + MobileNet v2 (Live Food Vision), Leaflet 1.9.4 + OpenStreetMap (Cubao GPS Locator), Open Food Facts Barcode API.
- Cloud & Backend Infrastructure: Node.js API + Serverless/Static Cloud Compatibility (Vercel, GitHub Pages, and HTTPS Cloud Tunneling).

E. Main Components

- 1. Smart Budget Meal Planner Engine (Low PHP 45-75, Medium PHP 80-135, High PHP 140-235 tiers + 7-Day Pro Planner).
- 2. Instant Cubao Karinderya GPS Locator (Leaflet interactive map + Haversine real-time distance calculation).
- 3. Real-Time AI Camera Food Scanner & Kidney Sodium Shield (Detects high-salt dialysis-risk meals and suggests healthy swaps).
- 4. Adaptive PC Holder & Mobile Phone Holder Switcher with Instant QR Code Handoff.

F. Safety Features (2+ Key Safety Considerations)

- 1. Chronic Kidney Disease (CKD) & Hypertension Sodium Safety Alerts: Automatically warns users when a scanned meal exceeds 850mg sodium or daily intake nears the 1,500mg-2,000mg DOH limit.
- 2. On-Device Camera Privacy & Secure Authentication: Processes live food camera frames inside browser memory without storing private photos on external servers, paired with

scrypt password hashing.

G. Sustainability Features (2+ Ways the Design Promotes Sustainability)

- 1. 100% Pure Digital Zero-Hardware Footprint: Requires zero plastic manufacturing or paper menus by running entirely on existing student smartphones and PCs.
- 2. Plant-Forward & Local Karinderya Sourcing (SDG 12): Shifts consumption away from plastic-wrapped instant noodles toward fresh local vegetables (monggo, malunggay, kangkong, tokwa) cooked by neighborhood Cubao karinderyas.

STEP 5: ESTIMATED COST, UNIT ECONOMICS & BREAK-EVEN ANALYSIS

Aligned with our Market Sizing (TAM: 600,000 | SAM: 25,000 | SOM: 1,000 active subscribers), below is our monthly unit economics and cost table for operating ByteBite at our 1,000-subscriber SOM target:

- Materials (Cloud database storage & nutrition dataset maintenance): PHP 4,000.00 / month
- Labor (Developer retainer & technical administration - Fixed): PHP 12,000.00 / month
- Manufacturing / Production (Cloud hosting servers & SSL infrastructure - Fixed): PHP 4,500.00 / month
- Packaging (Digital onboarding assets, QR kits & app store/web branding - Fixed): PHP 1,500.00 / month
- Transportation & Campus Marketing (Cubao karinderya partner visits & campus campaigns - Fixed): PHP 3,000.00 / month
- Other Costs / Variable Costs (PHP 19.00/user x 1,000 users: Cloud API queries, messaging support, payment gateway fees): PHP 19,000.00 / month

TOTAL MONTHLY COST AT 1,000 SUBSCRIBERS (PHP 25,000 Fixed + PHP 19,000 Variable): PHP 44,000.00 / month (Total Cost per Unit at 1,000 users = PHP 44.00 / user)

Estimated Selling Price (Monthly Subscription Fee): PHP 99.00 / user per month

Variable Cost per Subscription: PHP 19.00 / user (Contribution Margin = PHP 80.00 / user)

Break-Even Volume (BEP): 313 paid subscribers/month (PHP 25,000 Fixed Costs / PHP 80.00 Contribution Margin)

Projected Monthly Net Profit at 1,000 Subscribers (SOM): PHP 55,000.00 / month (PHP 99,000 Revenue - PHP 19,000 Variable Cost - PHP 25,000 Fixed Cost)

Pricing Explanation: At PHP 99.00/month (just PHP 3.30/day) and a low PHP 19.00 variable cost per user, each subscriber generates a PHP 80.00 Contribution Margin (80.8%). This allows ByteBite to cover its PHP 25,000 monthly fixed costs at just 313 subscribers and generate PHP 55,000.00 in monthly net profit at our 1,000-user SOM.

STEP 6: SDG IMPACT (SOCIAL, ENVIRONMENTAL, AND ECONOMIC VALUE)

1. Social Impact (How the solution helps people - SDG 3 & SDG 2)

Directly combats the rising prevalence of Chronic Kidney Disease (dialysis), diabetes, and hypertension among young Filipino adults by replacing high-sodium fast food with customized, kidney-safe meal plans and accessible Cubao karinderya options.

2. Environmental Impact (How it promotes sustainability - SDG 12 & SDG 13)

As a pure digital app that promotes freshly cooked local vegetable and fish dishes from neighborhood karinderyas over plastic-packaged instant noodles and fast-food takeout, ByteBite significantly reduces single-use plastic waste and food-supply carbon emissions.

3. Economic Impact (How it creates income and reduces costs - SDG 1 & SDG 8)

Saves students PHP 1,500-PHP 2,500/month on food, protects families from PHP 20,000+/month dialysis medical bills, drives daily customer revenue to local Cubao karinderyas, and builds

a profitable digital venture earning PHP 55,000/month net profit at 1,000 subscribers.

STEP 7: INDIVIDUAL 3-MINUTE PITCH SCRIPT (RENIER B. LOPEZ - IT41S7)

- 1. THE PROBLEM (0:00 - 0:30 | 30 Seconds): "Good day, Dr. Domingo and classmates! I am Renier B. Lopez from IT41S7, presenting ByteBite. Today, we face an alarming health problem: a high prevalence of lifestyle diseases-specifically Chronic Kidney Disease requiring dialysis, diabetes, and hypertension-among young Filipino adults. Because of busy schedules and tight budgets, students rely daily on cheap, high-sodium fast food and instant noodles that pack nearly 100% of the safe daily sodium limit in a single meal."
- 2. THE CUSTOMER & MARKET SIZE (0:30 - 0:50 | 20 Seconds): "Our target customers are budget-conscious university students and young adults. Our Total Addressable Market (TAM) is 600,000 students in Metro Manila, our Serviceable Available Market (SAM) is 25,000 students right here in Cubao, Quezon City, and our Serviceable Obtainable Market (SOM) is 1,000 active paid subscribers."
- 3. THE SOLUTION (0:50 - 1:35 | 45 Seconds): "Our solution is ByteBite-a pure digital Smart Healthy Meal Planner and Budget Locator App. First, it generates customized, kidney-safe Filipino meal plans across Low (PHP 45-75), Medium (PHP 80-135), and High (PHP 140-235) student budgets, complete with palengke ingredient costs and 12 Cooking Reels. Second, our interactive Leaflet GPS map instantly locates affordable healthy vendors and karinderyas in Cubao, QC. Third, our Real-Time AI Camera Scanner detects what you are eating-flagging high-sodium dialysis-risk meals and recommending healthy swaps."
- 4. THE LEAN CANVAS & UNIT ECONOMICS (1:35 - 2:35 | 60 Seconds): "In our Lean Canvas and Unit Economics, ByteBite Pro is priced at PHP 99.00 per user per month. Our variable cost is only PHP 19.00 per user (covering cloud API queries, messaging support, and payment gateway fees), giving us a strong Contribution Margin of PHP 80.00 per user. Against our Monthly Fixed Costs of PHP 25,000.00 (for cloud hosting, developer retainer, campus marketing, and admin fees), our Break-Even Volume is just 313 paid subscribers per month. At our target SOM of 1,000 active subscribers, ByteBite generates PHP 99,000.00 in monthly revenue, PHP 19,000.00 in total variable cost, and a Projected Monthly Net Profit of PHP 55,000.00."
- 5. PRODUCT DESIGN (2:35 - 2:55 | 20 Seconds): "As a pure digital app hosted on the cloud, ByteBite features a Dual-Holder Responsive Design: it runs as a full command dashboard on a PC Holder and switches with one click or QR scan into a Smartphone Mobile Holder."
- 6. SDG IMPACT (2:55 - 3:15 | 20 Seconds): "ByteBite directly advances SDG 3, SDG 2, and SDG 12: socially preventing kidney dialysis and hypertension, environmentally cutting plastic sachet waste, and economically boosting local Cubao karinderyas."
- 7. CLOSING (3:15 - 3:20 | 5 Seconds): "With ByteBite, eating on a student budget never has to cost you your kidneys. Thank you!"

STEP 8: SHORT REFLECTION (ANSWERS TO ALL 3 QUESTIONS)

1. What did you learn from developing your SDG-based innovation?

Developing ByteBite taught me that preventing lifestyle diseases like Chronic Kidney Disease, diabetes, and hypertension (SDG 3 and SDG 2) among students requires aligning nutrition directly with local student budgets. By mapping real Cubao karinderya and palengke prices, I learned that a pure digital app can connect students to PHP 45-PHP 75 freshly cooked Filipino meals that contain 80% less sodium than fast food. I also learned how clear unit economics can turn a student health advocacy into a profitable, self-sustaining digital

venture.

2. What was the most difficult part of developing your Lean Canvas?

The most challenging part of developing the Lean Canvas was structuring the Cost Structure, Unit Economics, and Break-Even point so that the subscription fee remains affordable for students while covering PHP 25,000 in monthly fixed costs. By optimizing our variable cost down to just PHP 19.00 per user using client-side AI and open-source mapping, we achieved a PHP 80.00 contribution margin on a PHP 99.00 monthly fee. That allowed us to reach break-even at a realistic 313 subscribers out of Cubao's 25,000 student SAM.

3. If you had more time and resources, what would you improve about your product/service?

If I had more time and resources, I would onboard 50+ additional karinderyas across Quezon City and the University Belt to expand our SAM toward our 600,000 Metro Manila TAM. I would also integrate direct GCash/Maya cashless student meal vouchers inside the app and partner with university clinics and the National Kidney and Transplant Institute (NKTI) for clinical nutrition validation.

STEP 9: DEFENSE ANSWERS TO ALL 10 PROBING QUESTIONS & REFERENCES

- Q1 (Problem): High prevalence of CKD/dialysis, diabetes, and hypertension among young adults because students rely on cheap, high-sodium fast food on tight budgets.
- Q2 (Customer): Cubao university students & young adults (TAM: 600,000 | SAM: 25,000 in Cubao, QC | SOM: 1,000 active subscribers).
- Q3 (UVP): A pure digital app hyper-localized to Cubao karinderyas, Low/Med/High PHP student budgets, real-time AI sodium scanning, and DOH kidney safety.
- Q4 (Solution): Generates customized budget meal plans (PHP 45-235), locates healthy Cubao karinderyas via live GPS map, and scans food via AI camera for sodium risk.
- Q5 (Channels): Cloud web/mobile hosting (Vercel/GitHub/QR link), campus partnerships at TIP QC, social cooking reels, and partner karinderya referrals.
- Q6 (Revenue): PHP 99.00/month subscription fee -> PHP 99,000.00 monthly revenue and PHP 55,000.00 net profit at 1,000 active subscribers.
- Q7 (Cost): PHP 19.00/user variable cost (PHP 80 contribution margin) and PHP 25,000.00 monthly fixed cost -> Break-Even Volume of 313 paid subscribers/month.
- Q8 (Metrics): Reaching 313 BEP and 1,000 SOM subscribers, keeping daily sodium <1,500mg-2,000mg, daily PHP savings, and karinderya GPS visits.
- Q9 (Unfair Advantage): Localized 1,009-meal Cubao karinderya/palengke dataset + ultra-low PHP 19/user variable cost via client-side Edge AI vision.
- Q10 (Integration): A pure digital model with an 80.8% contribution margin (PHP 80/user) lets us solve student health & budget pain (Problem/Customer/Solution) and break even at just 1.25% of our Cubao SAM (313 of 25,000 students).

10. References / Sources

- 1. Department of Health (DOH) & National Kidney and Transplant Institute (NKTI). Philippine Renal Disease Registry & CKD Prevention Guidelines.
- 2. World Health Organization (WHO). Guideline: Sodium intake for adults and children (<2,000 mg/day threshold).
- 3. Food and Nutrition Research Institute - DOST (FNRI-DOST). Philippine Food Composition Tables (FCT) & Pinggang Pinoy.
- 4. Open Food Facts & OpenStreetMap Contributors (2026). Quezon City Geospatial & Nutritional Data.